



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Name: Dharmit Kunal Shah

PRN: 25070122073

CSE A-2

Subject: ProgrammingwithJava

ASSIGNMENT No : 6

TITLE : Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling.

```
class Outer {  
    class Inner {  
        void display () {  
            System.out.println(x: "Inner Class Method");  
        }  
    }  
}  
  
public class Main {  
    Run | Debug  
    public static void main (String[] args) {  
        System.out.println(x: "Inner and Outer classes : ");  
        Outer.Inner obj = new Outer().new Inner();  
        obj.display();  
    }  
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp6> java Main.java  
Inner and Outer classes :  
Inner Class Method
```

Exercise

1. Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

```
interface Action {
    void start();
}

class Vehicle {
    String name;
    String model;

    Vehicle (String n, String m) {
        this.name = n;
        this.model = m;
    }

    class Display {
        void display() {
            System.out.println("Car Name : " + name);
            System.out.println("Car Model : " + model);
        }
    }
}

public class Ex1 {
    Run | Debug
    public static void main (String[] args) {
        Vehicle.Display v = new Vehicle(n: "Hyundai", m: "Exter").new Display();

        Action a = new Action () {
            @Override
            public void start () {
                System.out.println(x: "Car started!");
            }
        };

        v.display();
        a.start();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp6\Exercise> java Ex1.java
Car Name : Hyundai
Car Model : Exter
Car started!
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp6\Exercise> █
```

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

```

interface DeliveryStatus {
    void updateStatus();
}

class FoodDelivery {
    int orderId;
    String customerName;
    String foodItem;

    FoodDelivery (int orderId, String customerName, String foodItem) {
        this.orderId = orderId;
        this.customerName = customerName;
        this.foodItem = foodItem;
    }

    class OrderDetails {
        void display() {
            System.out.println("Order ID      : " + orderId);
            System.out.println("Customer Name : " + customerName);
            System.out.println("Food Item    : " + foodItem);
        }
    }
}

public class Ex2 {
    Run | Debug
    public static void main(String[] args) {
        FoodDelivery order = new FoodDelivery(orderId: 1058, customerName: "Dharmit", foodItem: "Manchurian");

        FoodDelivery.OrderDetails details = order.new OrderDetails();
        details.display();

        DeliveryStatus status = new DeliveryStatus() {
            @Override
            public void updateStatus() {
                System.out.println(x: "Delivery Status : Out for Delivery");
            }
        };

        status.updateStatus();
    }
}

```

```

Order ID      : 1058
Customer Name : Dharmit
Food Item     : Manchurian
Delivery Status : Out for Delivery

```